CSE4088 Introduction to Machine Learning

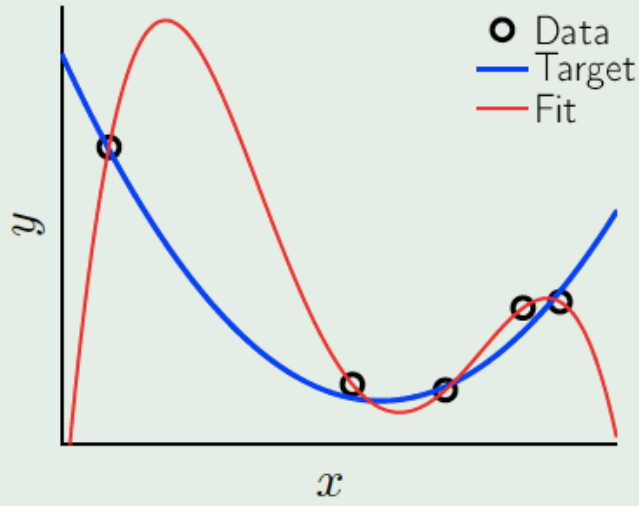
Regularization

Slides are adopted from lecture notes of Yaser Abu-Mostafa

Review of last lecture

- **Overfitting**

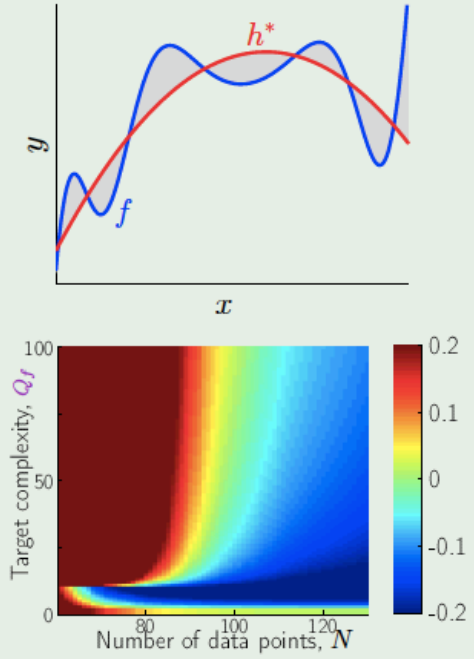
  Fitting the data more than is warranted



  VC allows it; doesn't predict it

Fitting the noise, stochastic/deterministic

- **Deterministic noise**



Outline

- Regularization – informal
- Regularization – formal
- Weight decay
- Choosing a regularizer

Two approaches to regularization

Mathematical:

Ill-posed problems in function approximation

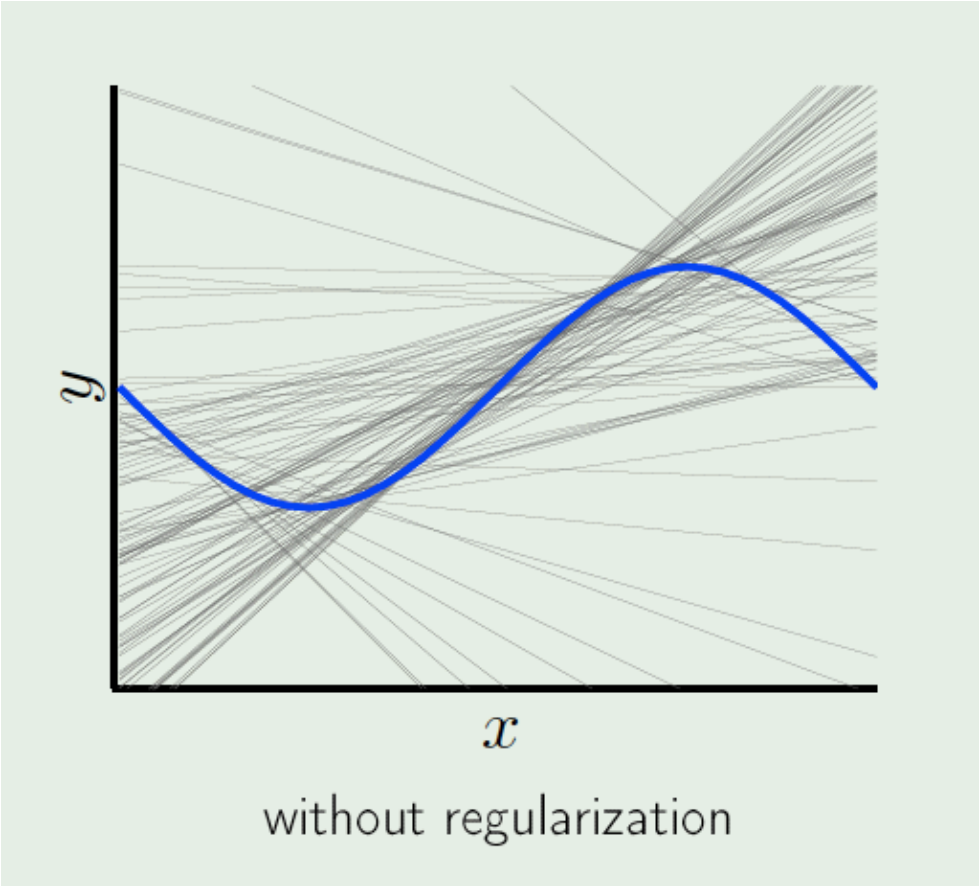
Heuristic:

Handicapping the minimization of E_{in}

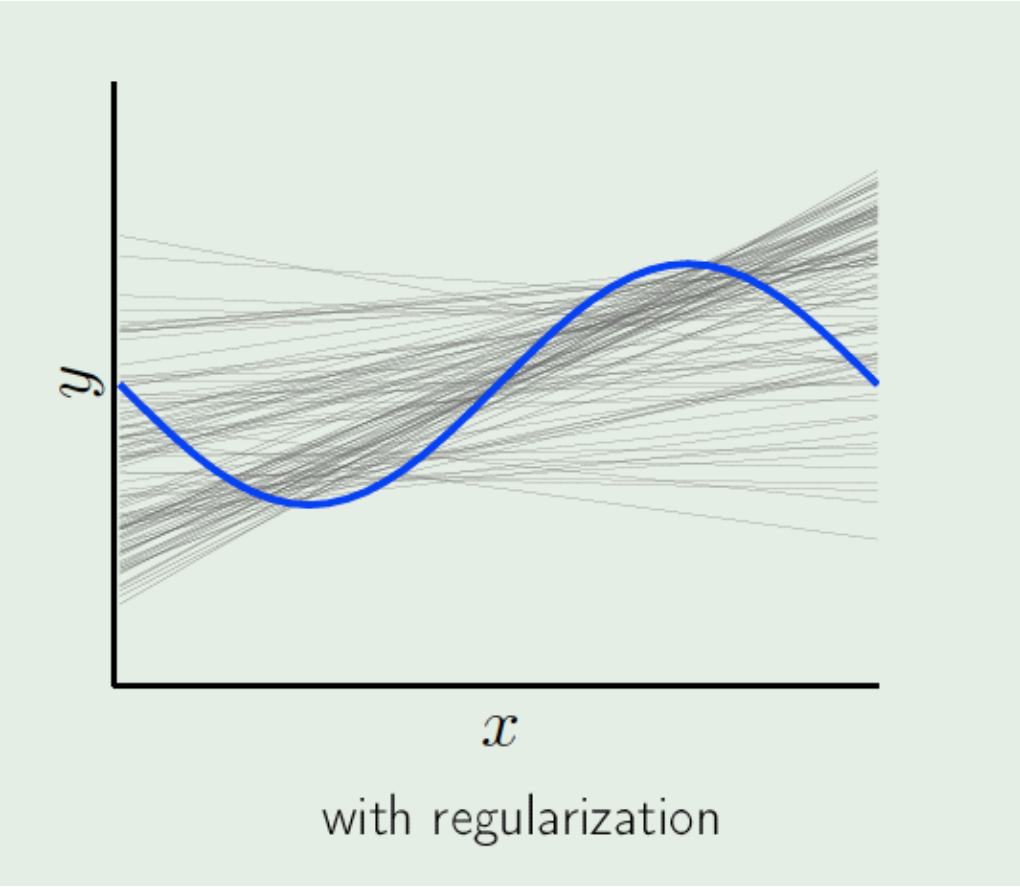
Mathematical: You want to approximate a function, but there are more than one solutions. Then, you impose smoothness constraints on it. Problem is that in most of the cases the assumptions made here are not realistic.

Heuristic: We will use mathematical-based approaches to fight overfitting.

A familiar example

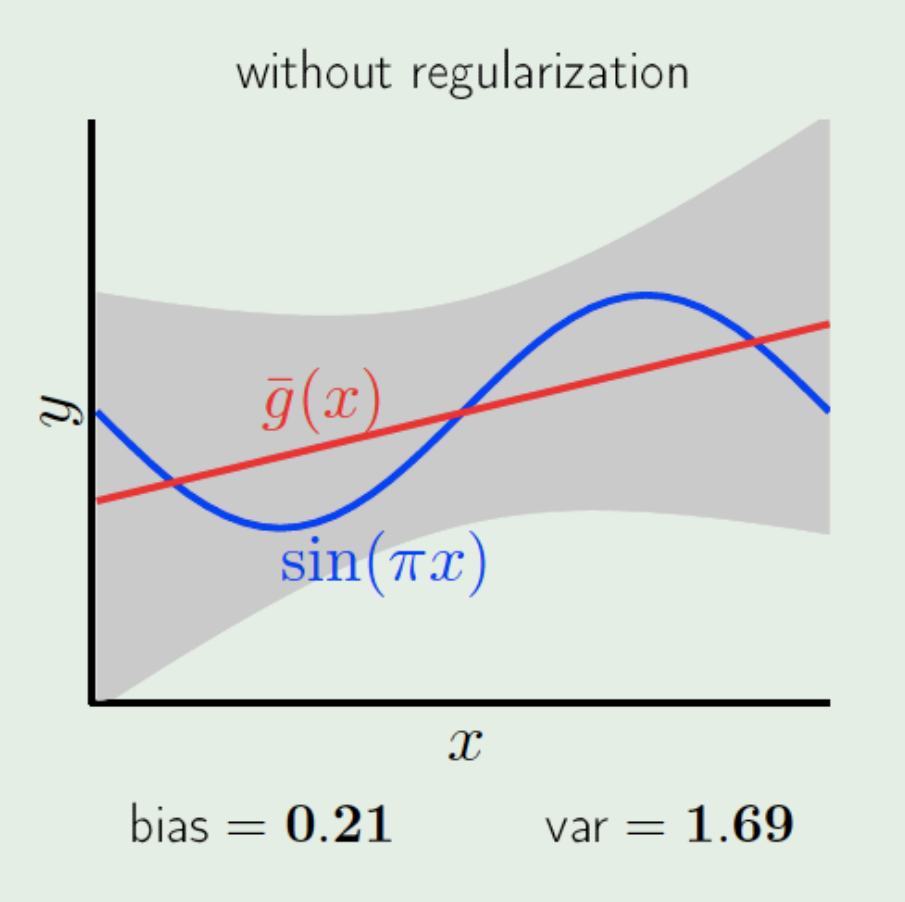


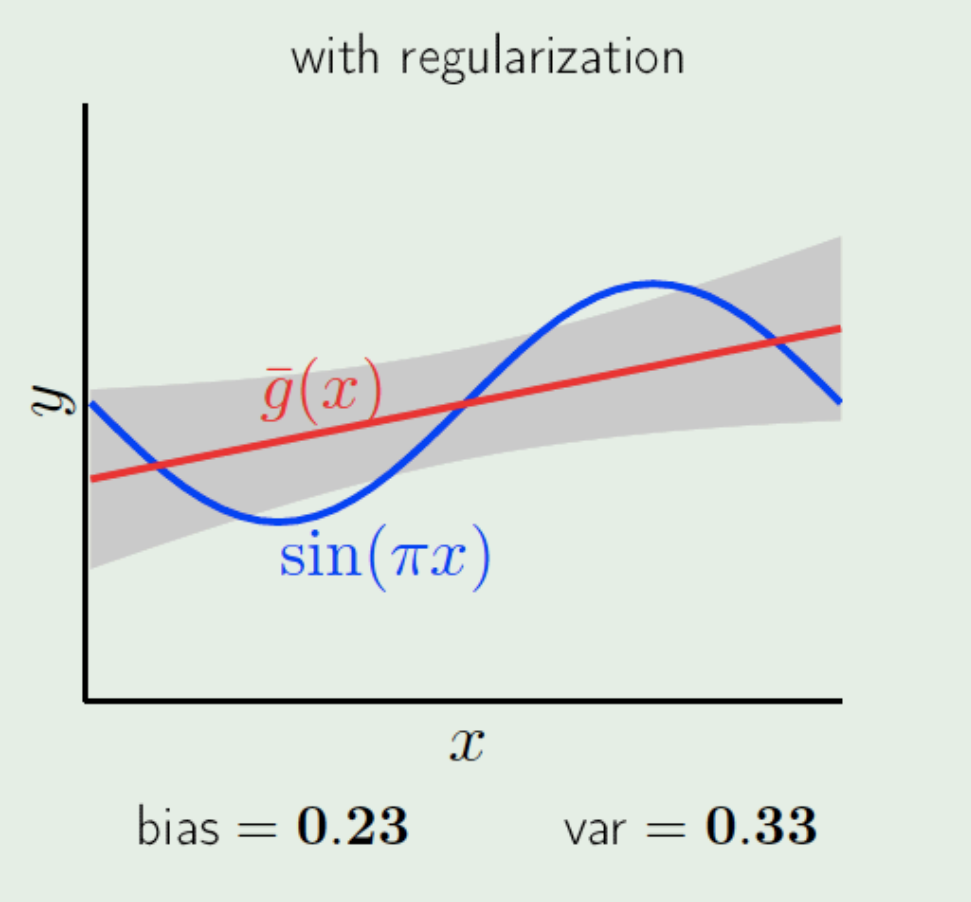
Given two points, the linear model has high variance.



The slope and offset of the line is regularized (constrained).

And the winner is …





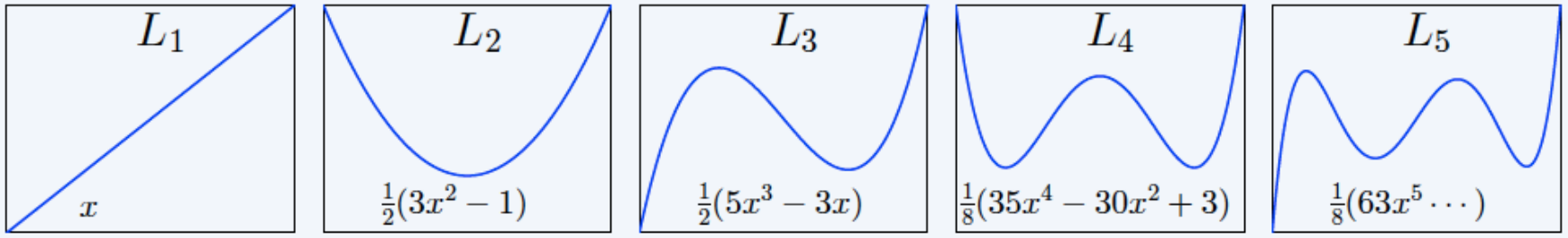
Regularization reduces var and increases bias a little bit.

The polynomial model

$\mathcal{H}_Q$: polynomials of order Q linear regression in $\mathcal{Z}$ space

$$\mathbf{z} = \begin{bmatrix} 1 \\ L_1(x) \\ \vdots \\ L_Q(x) \end{bmatrix} \qquad \mathcal{H}_Q = \left\{ \sum_{q=0}^{Q} w_q \, L_q(x) \right\}$$

Legendre polynomials:



Target function is unknown, which is the case in a learning scenario. We are going to use the parameterized hypothesis set to approximate our target function using a finite set.

Unconstrained solution

Given $(x_1, y_1), \cdots, (x_N, y_n) \longrightarrow (\mathbf{z}_1, y_1), \cdots, (\mathbf{z}_N, y_n)$

Minimize $E_{\text{in}}(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^{N} (\mathbf{w}^\intercal \mathbf{z}_n - y_n)^2$

Minimize $\frac{1}{N} (\mathrm{Z}\mathbf{w} - \mathbf{y})^\intercal (\mathrm{Z}\mathbf{w} - \mathbf{y})$

$\mathbf{w}_{\text{lin}} = (\mathrm{Z}^\intercal \mathrm{Z})^{-1} \mathrm{Z}^\intercal \mathbf{y}$

Constraining the weights

Hard constraint: $\mathcal{H}_2$ is constrained version of $\mathcal{H}_{10}$ with $w_q = 0$ for $q > 2$

Softer version: $\sum_{q=0}^{Q} w_q^2 \leq C$ "soft-order" constraint

Minimize $\frac{1}{N}(\mathbf{Z}\mathbf{w} - \mathbf{y})^{\mathsf{T}}(\mathbf{Z}\mathbf{w} - \mathbf{y})$

subject to: $\mathbf{w}^{\mathsf{T}}\mathbf{w} \leq C$

Solution: $\mathbf{w}_{\text{reg}}$ instead of $\mathbf{w}_{\text{lin}}$

Solving for $\mathbf{w}_{\text{reg}}$

Minimize $E_{\text{in}}(\mathbf{w}) = \frac{1}{N}(\mathbf{Z}\mathbf{w} - \mathbf{y})^{\mathsf{T}}(\mathbf{Z}\mathbf{w} - \mathbf{y})$

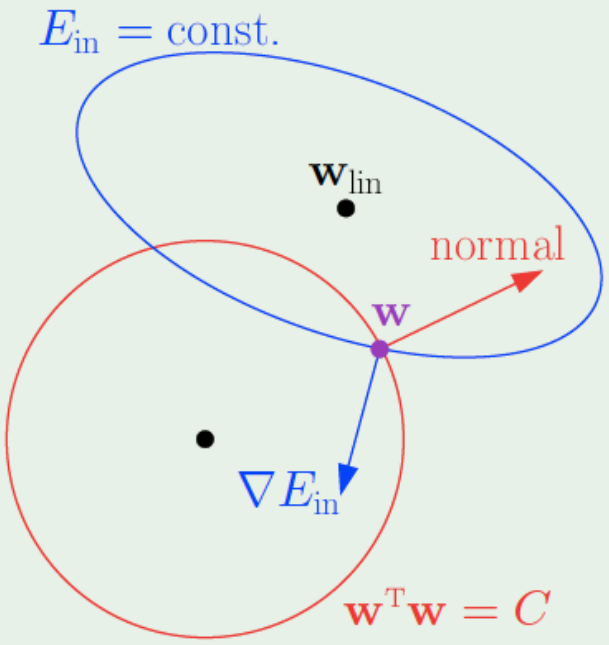
subject to: $\mathbf{w}^{\mathsf{T}}\mathbf{w} \leq C$

$$\nabla E_{\text{in}}(\mathbf{w}_{\text{reg}}) \propto -\mathbf{w}_{\text{reg}}$$

$$= -2\tfrac{\lambda}{N}\mathbf{w}_{\text{reg}}$$

$$\nabla E_{\text{in}}(\mathbf{w}_{\text{reg}}) + 2\tfrac{\lambda}{N}\mathbf{w}_{\text{reg}} = \mathbf{0}$$

Minimize $E_{\text{in}}(\mathbf{w}) + \frac{\lambda}{N}\mathbf{w}^{\mathsf{T}}\mathbf{w}$ $\boxed{C \uparrow \quad \lambda \downarrow}$



Augmented error

Minimizing $E_{\text{aug}}(\mathbf{w}) = E_{\text{in}}(\mathbf{w}) + \frac{\lambda}{N}\mathbf{w}^T\mathbf{w}$

$= \frac{1}{N}(\mathbf{Z}\mathbf{w} - \mathbf{y})^T(\mathbf{Z}\mathbf{w} - \mathbf{y}) + \frac{\lambda}{N}\mathbf{w}^T\mathbf{w}$ unconditionally

– solves –

Minimizing $E_{\text{in}}(\mathbf{w}) = \frac{1}{N}(\mathbf{Z}\mathbf{w} - \mathbf{y})^T(\mathbf{Z}\mathbf{w} - \mathbf{y})$

subject to: $\mathbf{w}^T\mathbf{w} \leq C$ ⟵ VC formulation

Since some hypothesis are not allowed, we expect the VC dimension to decrease and hence generalization behavior will improve.

The solution

Minimize $E_{\text{aug}}(\mathbf{w}) = E_{\text{in}}(\mathbf{w}) + \frac{\lambda}{N}\mathbf{w}^{\mathsf{T}}\mathbf{w}$

$$= \frac{1}{N}\left((\mathrm{Z}\mathbf{w} - \mathbf{y})^{\mathsf{T}}(\mathrm{Z}\mathbf{w} - \mathbf{y}) + \lambda\, \mathbf{w}^{\mathrm{T}}\mathbf{w}\right)$$

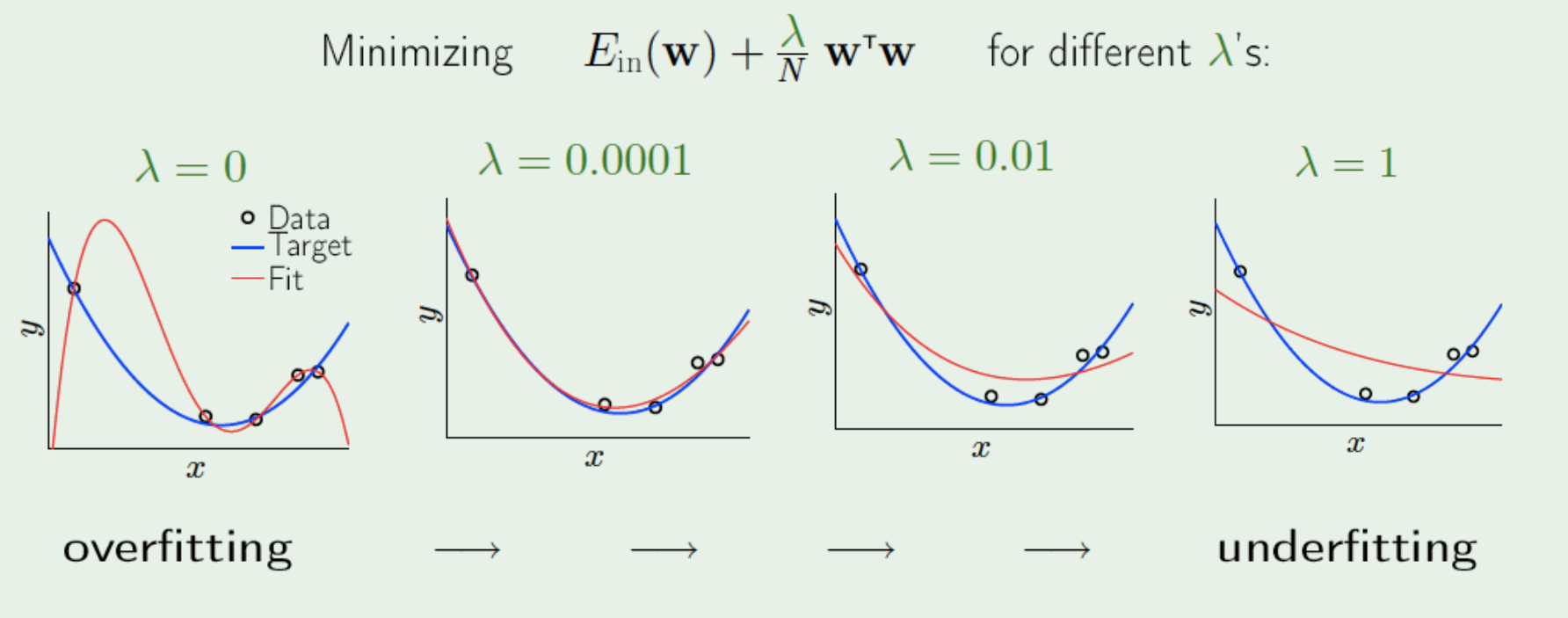
$$\nabla E_{\text{aug}}(\mathbf{w}) = \mathbf{0} \quad \Longrightarrow \quad \mathrm{Z}^{\mathsf{T}}(\mathrm{Z}\mathbf{w} - \mathbf{y}) + \lambda\mathbf{w} = \mathbf{0}$$

$$\boxed{\mathbf{w}_{\text{reg}} = (\mathrm{Z}^{\mathsf{T}}\mathrm{Z} + \lambda\mathrm{I})^{-1}\mathrm{Z}^{\mathsf{T}}\mathbf{y}} \qquad \text{(with regularization)}$$

as opposed to $\mathbf{w}_{\text{lin}} = (\mathrm{Z}^{\mathsf{T}}\mathrm{Z})^{-1}\mathrm{Z}^{\mathsf{T}}\mathbf{y}$ (without regularization)

If λ is large, then we wil have $\frac{1}{\lambda}$ on the right hand side and w will be pıshed towards zero → strong regularization.

The result


Minimizing $E_{\text{in}}(\mathbf{w}) + \frac{\lambda}{N}\mathbf{w}^{\mathsf{T}}\mathbf{w}$ for different λ's:
$\lambda = 0$
$\lambda = 0.0001$
$\lambda = 0.01$
$\lambda = 1$
Data
Target
Fit
y
x
overfitting
underfitting

Weight 'decay'

Minimizing $E_{\text{in}}(\mathbf{w}) + \frac{\lambda}{N}\,\mathbf{w}^{\mathsf{T}}\mathbf{w}$ is called weight *decay*. Why?

Gradient descent:

$$\mathbf{w}(t+1) = \mathbf{w}(t) \;-\; \eta\,\nabla E_{\text{in}}\left(\mathbf{w}(t)\right) \;-\; 2\,\eta\,\frac{\lambda}{N}\,\mathbf{w}(t)$$

$$= \mathbf{w}(t)\,\left(1 - 2\eta\frac{\lambda}{N}\right) - \eta\,\nabla E_{\text{in}}\left(\mathbf{w}(t)\right)$$

Shrinks the weight (e.g. 0.99)

Applies in neural networks:

$$\mathbf{w}^{\mathsf{T}}\mathbf{w} = \sum_{l=1}^{L}\sum_{i=0}^{d^{(l-1)}}\sum_{j=1}^{d^{(l)}}\left(w_{ij}^{(l)}\right)^2$$

Variations of weight decay

Emphasis of certain weights: $\sum_{q=0}^{Q} \gamma_q \, w_q^2$ (if γ is small we can make that weight big)

Examples:

$\gamma_q = 2^q \implies$ low-order fit

$\gamma_q = 2^{-q} \implies$ high-order fit

Neural networks: different layers get different γ's

Tikhonov regularizer: $\mathbf{w}^\intercal \Gamma^\intercal \Gamma \mathbf{w}$

Even weight growth!

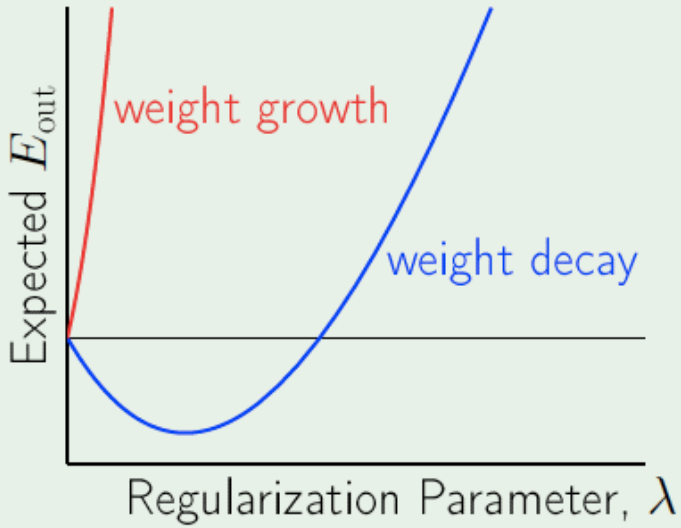
We 'constrain' the weights to be large - bad!

Practical rule:

stochastic noise is 'high-frequency'

deterministic noise is also non-smooth

$\Longrightarrow$ constrain learning towards smoother hypotheses



Since noise is high-frequency, we want to select a hypothesis that is smooth (e.g. small weights), which harms overfitting the noise more than it harms the fitting the signal.

General form of augmented error

Calling the regularizer $\Omega = \Omega(h)$, we minimize

$$E_{\text{aug}}(h) = E_{\text{in}}(h) + \frac{\lambda}{N}\Omega(h)$$

Rings a bell? $\downarrow\downarrow$

$$E_{\text{out}}(h) \leq E_{\text{in}}(h) + \Omega(\mathcal{H})$$

E_{aug} is better than E_{in} as a proxy for E_{out}

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- Regularization – formal
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The perfect regularizer Ω

Constraint in the 'direction' of the target function (going in circles ☺)

Guiding principle:

Direction of **smoother** or "simpler"

Chose a bad Ω?

We still have λ!

Guideline: We do not know the target function Regularization is a general methodology that harms fitting to noise more than it harms fitting to the signal. We move in the direction of smoother hypothesis, since noise is not smooth.

Neural-network regularizers

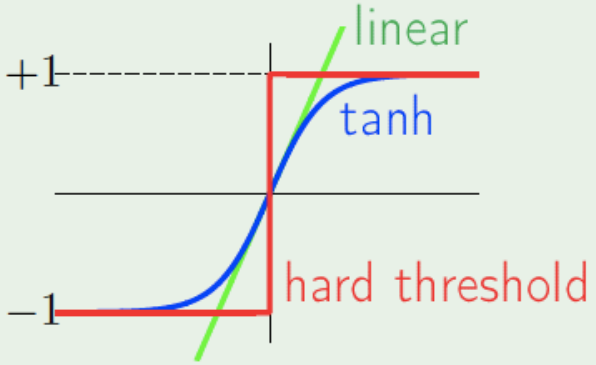
Weight decay: From linear to logical

Weight elimination:

Fewer weights $\Longrightarrow$ smaller VC dimension

Soft weight elimination:

$$\Omega(\mathbf{w}) = \sum_{i,j,l} \frac{\left(w_{ij}^{(l)}\right)^2}{\beta^2 + \left(w_{ij}^{(l)}\right)^2}$$



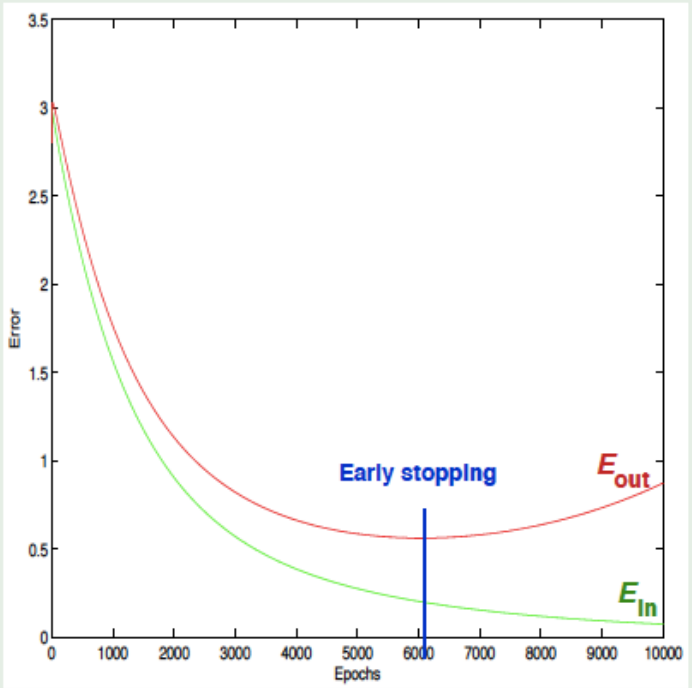
--If you force all weights to be Small, your big network just implements a linear function.

-- Hence emphasize some Weights and kill the others, to operate in the nonlinear region: if w is small beta will dominate and weight will decay, if w is large, do not change it.

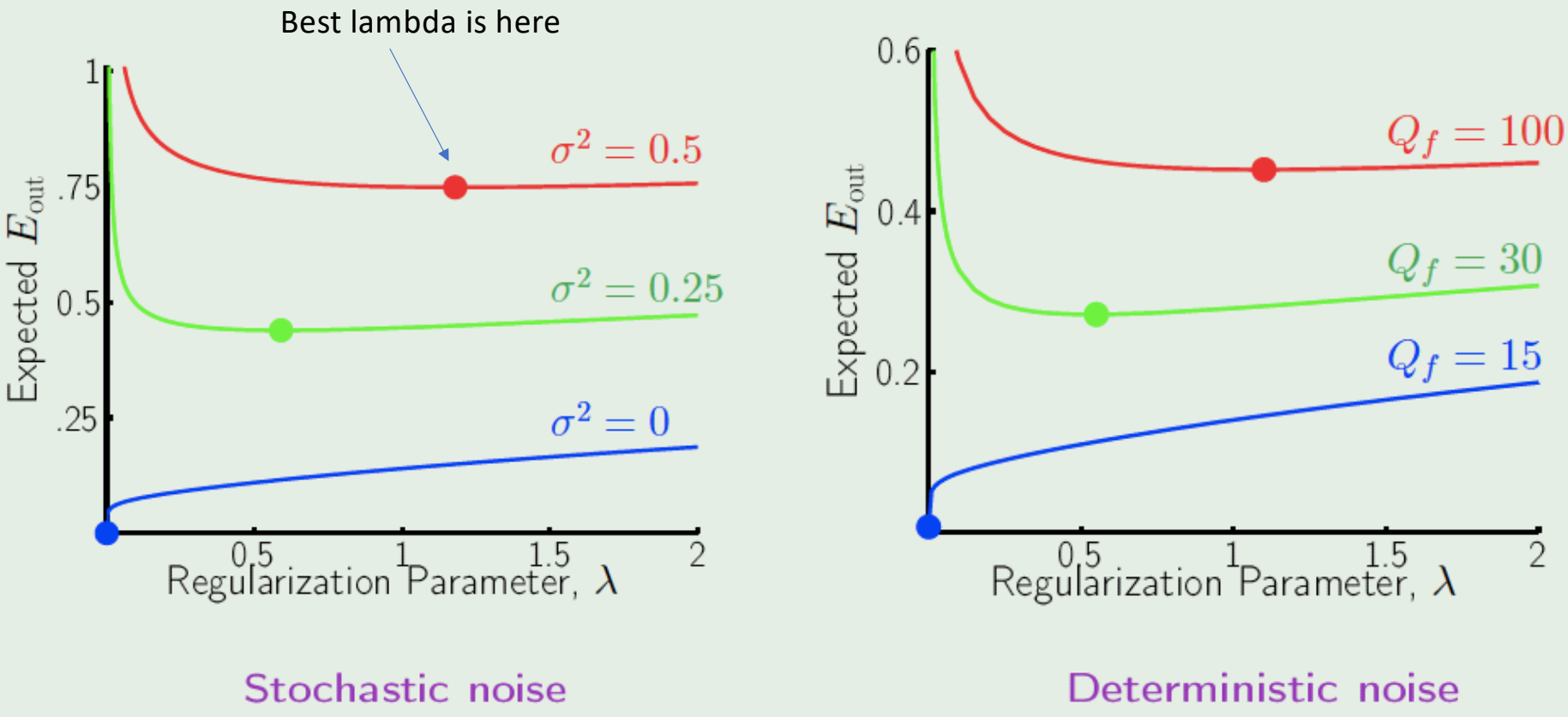
Early stopping as a regularizer

Regularization through the optimizer!

When to stop? **validation**



The optimal λ



As far as overftting and its cures are concerned, deterministic noise behaves exactly as unknown stochastic noise.